Homework 7

Graded

Student

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Total Points

60 / 60 pts

Question 1

Problem 1

10 / 10 pts

- **− 2 pts** $E(X)$
- **− 2 pts** $E(X^2)$
- **− 2 pts** $Var(X)$
- **− 2 pts** $P(\bar{X} < 1.6) = P(Z < \frac{\sqrt{15}}{5})$
- **− 2 pts** answer

Click here to replace this description.

- ✔ **− 0 pts** Correct
- **− 10 pts** Incorrect

Question 2

Problem 2

20 / 20 pts

✔ **– 0 pts** Correct

– 2 pts (a) $X \sim NBin(r = 100, p = 1/4)$

– 2 pts (a) $E(X)$

– 2 pts (a) $Var(X)$

– 2 pts (a) Using continuity correction

– 2 pts (a) answer

Click here to replace this description.

– 2 pts (b) $E(\epsilon_i)$

– 2 pts (b) $Var(\epsilon_i)$

– 2 pts (b) $X - Y \simeq N(0, 625)$

– 2 pts (b) $P(|\sum \epsilon_i| > 50) = P(\sum \epsilon_i > 50) + P(\sum \epsilon_i < -50)$

– 2 pts (b) answer

– 20 pts Incorrect

Question 3

Problem 3

10 / 10 pts

✔ **– 0 pts** Correct

– 2 pts forgot a factor of n in the final answer

– 1 pt forgot to subtract final answer by 1

– 1 pt fraction off by a factor of negative 1

– 1 pt answer has ln of inverse of x not just the ln of x

– 1 pt ln should be in the denominator

– 2 pts incorrect log likelihood

– 1 pt some other algebra error

– 10 pts problem missing

– 2 pts derivative taken incorrectly

Question 4

Problem 4

20 / 20 pts

✔ **− 0 pts** Correct

− 3 pts (a) incorrect likelihood function and product form of PMF

− 3 pts (a) incorrect log-likelihood expression

− 3 pts (a) incorrect derivative of log-likelihood

− 3 pts (a) incorrect steps to isolate p

− 2 pts (a) incorrect final MLE, should be $\hat{p}_{\text{MLE}} = \frac{1}{\bar{x}}$ or $\frac{n}{\sum x_i}$

− 2 pts (b) should state population mean $\mathbb{E}[X] = \frac{1}{p}$

− 2 pts (b) should set sample mean equal to population mean

− 2 pts (b) correct final MME should be $\hat{p}_{\text{MME}} = \frac{1}{\bar{x}}$

Question assigned to the following page: 1

$36 - 12x + x^2$

$(6-x)(6-x)$

Homework 7

$n = 81$

1. $X_1, X_2, \ldots\ldots X_{81}$

$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$

$$f(x) = \begin{cases} \frac{(6-x)^2}{72} & 0 < x < 6 \\ 0 & \text{otherwise} \end{cases}$$

$P(\bar{X} < 1.6)$

$$\mu = E[x] = \int_{-\infty}^{\infty} x f(x)\, dx = \int_0^6 x(6-x)^2 \, dx$$

$$= \int_0^6 x(x^2 - 12x + 36)\, dx = \int_0^6 x^3 - 12x^2 + 36x \, dx$$

$$= \frac{x^4}{4} - \frac{12x^3}{3} + \frac{36x^2}{2} \Big|_0^6 = \frac{1296}{4} - 864 + 648 = 108$$

$$\frac{1}{72} \cdot \frac{108}{1} = 1.5$$

$$E[x^2] = \int_{-\infty}^{\infty} x^2 f(x)\, dx = \int_0^6 x^2 (6-x)^2 \, dx$$

$$= \int_0^6 x^2(x^2 - 12x + 36)\, dx = \int_0^6 x^4 - 12x^3 + 36x^2 \, dx$$

$$= \frac{x^5}{5} - \frac{12x^4}{4} + \frac{36x^3}{3} \Big|_0^6 = \frac{7776}{5} - 3888 + 2592$$

$$= \frac{259.2}{72} = 3.6$$

Question assigned to the following page: 1

$\sigma = \sqrt{\frac{1.35}{81}}$

$\frac{x - \mu}{\sigma} = z$

$\bar{X} \sim N\left(1.5, \frac{1.35}{81}\right)$

$E[\bar{x}] = E[x] = 1.5$

$Var(\bar{x}) = \frac{1.35}{81}$

How? 1. Com

$\sigma^2 = E[x^2] - (E[x])^2 = 3.6 - (1.5)^2$

$= 3.6 - 2.25 = 1.35$

$P(\bar{x} < 1.6) = P\left(z < \frac{1.6 - 1.5}{\sqrt{\frac{1.35}{81}}}\right)$

$= P(z < 0.775) = \boxed{\Phi(0.775)}$

0.78 using standard normal table

or

$\Phi\left(\frac{0.1}{\sqrt{1.35/81}}\right) = \Phi\left(\frac{0.1 \cdot \sqrt{81}}{\sqrt{1.35}}\right) = \boxed{\Phi\left(\frac{0.9}{\sqrt{1.35}}\right)}$

Question assigned to the following page: 2

$p = \frac{1}{4}$), $E[x] = \frac{r}{p}$ → 400, $Var(x) = \frac{r(1-p)}{p^2}$ → 1200

By the CLT: $X \approx N(\mu = 400, \sigma^2 = 1200)$

continuity correction

$P(x \geq 1000) = P(X \geq 999.5) = 1 - P(999.5) = 1 - P\left(\frac{999.5 - 400}{\sqrt{1200}}\right) = 1 - P\left(\frac{599.5}{\sqrt{1200}}\right)$ → 17.31

$\boxed{= 1 - \Phi\left(\frac{599.5}{\sqrt{1200}}\right) = 1 - 1 = 0}$ → Extremely large z-value would be far in the tail, making it one.

b) $\epsilon_1, \ldots, \epsilon_n \in (-0.5, 0.5)$, $n = 7500$

$P\left(Z > \frac{50}{\sqrt{625}}\right) = P\left(Z < \frac{50}{\sqrt{625}}\right)$ comes 2P, standard normal symmetric about z=0

$E[\epsilon_i] = \frac{-0.5 + 0.5}{2} = 0$, $Var[\epsilon_i] = \frac{(0.5 - (-0.5))^2}{12} = \frac{1}{12}$

$X - Y = \epsilon = \sum_{i=1}^{7500} \epsilon_i$, $E[\epsilon] = 0$ and $Var(\epsilon) = \frac{1}{12} \cdot 7500 = 625$

By the CLT, $\epsilon \approx N(\mu = 0, \sigma^2 = 625)$ $2(1 - \Phi(2))$ $\boxed{2(1 - \Phi(2))}$

$P(|\epsilon| > 50) = P(\epsilon > 50) + P(\epsilon < -50)$

$= P\left(\frac{\epsilon - 0}{\sqrt{625}} > \frac{50 - 0}{\sqrt{625}}\right) + P\left(\frac{\epsilon - 0}{\sqrt{625}} < -\frac{50 - 0}{\sqrt{625}}\right) = 2P(z > 2) = 2(1 - P(z \leq 2)) = 0.0455$

Question assigned to the following page: 3

HW7 Con.

3.

$$f_\theta(x) = \begin{cases} (\theta+1)x^\theta & 0<x<1 \\ 0 & \text{otherwise} \end{cases}$$

$$L(\theta) = \prod_{i=1}^{n} f_\theta(x_i) = \prod_{i=1}^{n} (\theta+1)x_i^\theta = (\theta+1)^n \cdot \prod_{i=1}^{n} x_i^\theta$$

$$L(\theta) = (\theta+1)^n \cdot \left(\prod_{i=1}^{n} x_i\right)^\theta$$

log turns multiplication ($\prod$) to sum (Σ).

$$\ell(\theta) = \ln L(\theta) = n\ln(\theta+1) + \theta\sum_{i=1}^{n} \ln x_i$$

$$\ell'(\theta) = \frac{n}{\theta+1} + \sum_{i=1}^{n} \ln x_i$$

$$\frac{n}{\theta+1} + \sum_{i=1}^{n} \ln x_i = 0$$

$$\frac{n}{\theta+1} = -\sum_{i=1}^{n} \ln x_i$$

$$\theta+1 = -\frac{n}{\sum_{i=1}^{n} \ln x_i}$$

$$\boxed{\hat{\theta} = -\frac{n}{\sum_{i=1}^{n} \ln x_i} - 1}$$

$$\ell''(\theta) = -\frac{n}{(\theta+1)^2}$$

, $n>0$ and $(\theta+1)^2>0$ so second derivative is negative, meaning that the function is concave down and the critical point is a max.

Multiplication of terms turns into adding terms or summing when log is used)

$$\ln\left(\prod_{i=1}^{n} a_i\right) = \sum_{i=1}^{n} \ln a_i$$

$$\ln(a \cdot b \cdot c) = \ln a + \ln b + \ln c + \ldots$$

$$\ln L(\theta) = \ln\left(\prod_{i=1}^{n} f_\theta(x_i)\right) = \sum_{i=1}^{n} \ln f_\theta(x_i)$$

Question assigned to the following page: 4

$\cdot\, p$ $\qquad \prod_{i=1}^{n} p = p^n,\ \prod_{i=1}^{n} a^{b_i} = a^{\sum b_i}$

$L(p) = \prod_{i=1}^{n} (1-p)^{x_i - 1} \cdot p = \prod_{i=1}^{n} \left[(1-p)^{x_i - 1} \cdot p \right]$

$L(p) = \left(\prod_{i=1}^{n} (1-p)^{x_i - 1} \right) \cdot \left(\prod_{i=1}^{n} p \right) = (1-p)^{\sum_{i=1}^{n} (x_i - 1)} \cdot p^n$

HW7 con.

Turn into summation b/c multiplication of exponents is addition!

4. $P(X=x) = (1-p)^{x-1} p \quad , x = 1, 2, 3$

a) $L(p) = \prod_{i=1}^{n} P(X_i = x_i) = \prod_{i=1}^{n} (1-p)^{x_i - 1} p = p^n (1-p)^{\sum_{i=1}^{n} (x_i - 1)}$

$$L(p) = p^n (1-p)^{\sum x_i - 1}$$

$$\ell(p) = \ln L(p) = n \ln p + \left(\sum_{i=1}^{n} (x_i - 1) \right) \ln(1-p)$$

$$\ell'(p) = \frac{n}{p} - \frac{\sum_{i=1}^{n} (x_i - 1)}{1-p} = 0$$

$$\frac{n}{p} = \frac{\sum_{i=1}^{n} (x_i - 1)}{1-p} \quad \rightarrow \text{cross multiply}$$

$$n(1-p) = p \sum_{i=1}^{n} (x_i - 1)$$

$$n - np = p \left(\sum_{i=1}^{n} x_i - n \right)$$

$$n - np = p \sum_{i=1}^{n} x_i - np$$

$$n = p \sum_{i=1}^{n} x_i$$

$$\boxed{\hat{p} = \frac{n}{\sum_{i=1}^{n} x_i}} \rightarrow \text{MLE}$$

$$\ell''(p) = -\frac{n}{p^2} - \frac{\sum_{i=1}^{n} (x_i - 1)}{(1-p)^2} < 0 \rightarrow \text{Confirms max, concave down, negative 2nd deriv.}$$

$E[x] = \frac{1}{p}$

$\bar{X} = \frac{1}{n} \sum_{i=1}^{n} x_i$

$\frac{1}{p} = \bar{X} \rightarrow p = \frac{1}{\bar{x}}$] → Equating sample mean to population mean → $\boxed{p = \frac{1}{\frac{1}{n}\sum_{i=1}^{n} x_i} = \frac{n}{\sum_{i=1}^{n} x_i}}$

↓

MME

MLE = MME, consistent estimator converging to the true p as $n \to \infty$, and unbiased so expected value equals p

• Note: $\sum_{i=1}^{n} (x_i - 1) = \sum x_i - n$ → These terms are equivalent

↳ Sum from 1 to n applies to one too, adding 1 a total of n times, which equals n.

Repeating the constant a certain number of times!